

Projective Puiseux Certificates for Unboundedness in Polynomial Optimization

Draft solution note

June 2026

Abstract

Nie and Yang asked for a computationally convenient certificate of unboundedness for a polynomial optimization problem when their horizon moment-ray condition is infeasible. This note gives a precise answer at the level of exact real-algebraic certificates. The certificate is a projective Puiseux arc along which the homogenized objective has a first negative term of order strictly smaller than the order forced by homogenization. The certificate is sound and complete for unboundedness below on a basic closed semialgebraic set. A Fritz–John tangency variant is also proved; it restricts the search to branches of an explicit polynomial system. The result should be read as substantial partial progress on Nie–Yang’s computational question rather than as a complete SDP-style solution: it gives a finite, exact, and checkable branch certificate, but it does not by itself provide a polynomial-time algorithm or a single moment/SOS relaxation whose infeasibility or feasibility is the desired certificate.

1 Introduction and status of the result

Let

$$g_* := \inf\{g(x) : x \in K\}, \quad K = \{x \in \mathbb{R}^n : c_i(x) = 0 \ (i \in \mathcal{E}), \ c_j(x) \geq 0 \ (j \in \mathcal{I})\}, \quad (1)$$

where all polynomials are real and where $\mathcal{E}, \mathcal{I}$ are finite. In Appendix A of [4], Nie and Yang study certificates for the assertion $g_* = -\infty$. Their horizon certificate is a moment-ray certificate supported on the set of feasible directions at infinity. They show that this certificate is sufficient but not necessary, and ask what should replace it when it is infeasible.

This note proves the following statement.

There is a sound and complete exact certificate for $g_ = -\infty$, namely a projective Puiseux descent arc. Moreover, if $g_* = -\infty$, one may choose such an arc from the Fritz–John tangency system obtained by minimizing g on large spheres.*

The word “exact” is important. The certificate below is a rigorous mathematical certificate and can be checked by real-algebraic computations such as Puiseux expansion, rational univariate representations, Thom encodings, and univariate sign determination. It is not, in the present form, a replacement for the moment-ray certificate by an equally simple SDP feasibility problem. Thus, relative to the literal phrase “computationally convenient” in [4, Question 7.1], the contribution should be classified as *partial progress*: it solves the missing higher-order certificate problem in exact algebraic terms, but it leaves open the stronger problem of finding an efficient or purely moment/SOS certificate with comparable convenience to the original horizon test.

2 Homogenization and the horizon moment ray

Let $d = \deg g$. We exclude the trivial case $d = 0$, since a constant polynomial cannot be unbounded below. For a polynomial p of degree e , write

$$\tilde{p}(x_0, x) = x_0^e p(x/x_0)$$

for its homogenization to degree e , and write $p_{\text{hom}}(x) = \tilde{p}(0, x)$ for its top homogeneous part. Define the compact projective feasible set

$$\widetilde{K} = \left\{ (x_0, x) \in \mathbb{R}^{n+1} : \begin{array}{ll} \tilde{c}_i(x_0, x) = 0 & (i \in \mathcal{E}), \\ x_0^2 + \|x\|^2 = 1, & x_0 \geq 0 \end{array} \quad \tilde{c}_j(x_0, x) \geq 0 \quad (j \in \mathcal{I}), \right\}. \quad (2)$$

The horizon set used by the ray certificate is

$$K^\infty = \left\{ u \in \mathbb{R}^n : \begin{array}{ll} (c_i)_{\text{hom}}(u) = 0 & (i \in \mathcal{E}), \\ \|u\| = 1 \end{array} \quad (c_j)_{\text{hom}}(u) \geq 0 \quad (j \in \mathcal{I}), \right\}. \quad (3)$$

This is the slice $\widetilde{K} \cap \{x_0 = 0\}$, written in the affine coordinate $u = x$.

Let $R_d(K^\infty)$ denote the degree- d truncated moment cone of finite positive Borel measures on K^∞ :

$$R_d(K^\infty) = \left\{ \left(\int_{K^\infty} u^\alpha d\mu(u) \right)_{|\alpha| \leq d} : \mu \geq 0 \text{ finite on } K^\infty \right\}.$$

The Nie–Yang horizon moment-ray system has the form

$$\langle g_{\text{hom}}, z \rangle = -1, \quad z \in R_d(K^\infty). \quad (4)$$

Proposition 2.1 (What the horizon ray detects). *The system (4) is feasible if and only if there is a point $u \in K^\infty$ such that $g_{\text{hom}}(u) < 0$.*

Proof. If $z \in R_d(K^\infty)$ is represented by a positive measure μ , then

$$\langle g_{\text{hom}}, z \rangle = \int_{K^\infty} g_{\text{hom}}(u) d\mu(u).$$

If this integral equals -1 , then g_{hom} must be negative at some point of the support of μ . Conversely, if $g_{\text{hom}}(u) < 0$ for some $u \in K^\infty$, then the measure

$$\mu = \frac{1}{-g_{\text{hom}}(u)} \delta_u$$

represents a moment vector satisfying (4). □

Thus (4) only sees a negative order-zero term of the homogenized objective at infinity. If $g_{\text{hom}} \geq 0$ on K^∞ , then a proof of unboundedness must use a branch approaching a zero of g_{hom} and detect a negative lower-order term along that branch. The projective Puiseux certificate below is designed for exactly this situation.

3 Puiseux germs and curve selection at infinity

We use $\mathcal{P}$ for the ordered field of real convergent Puiseux-Laurent germs at 0^+ , i.e. finite Laurent tails followed by a convergent series in some fractional power $t^{1/q}$. Every nonzero $h \in \mathcal{P}$ has a leading expansion

$$h(t) = at^r + O(t^{r+\epsilon}), \quad a \in \mathbb{R} \setminus \{0\}, \quad r \in \mathbb{Q},$$

and we write $\text{ord } h = r$, $\text{lc } h = a$. Inequalities in $\mathcal{P}$ are interpreted for all sufficiently small positive t , equivalently by the sign of the leading coefficient. We also put $\text{ord } 0 = +\infty$.

The following standard fact is used repeatedly.

Lemma 3.1 (Puiseux curve selection at infinity). *Let $S \subset \mathbb{R}^N$ be semialgebraic. If there are points of S escaping to infinity, then there is a semialgebraic arc $\gamma : (0, \epsilon) \rightarrow S$ such that $\|\gamma(t)\| \rightarrow \infty$ as $t \downarrow 0$. After replacing t by a positive fractional power and shrinking ϵ , every component of γ is represented by an element of $\mathcal{P}$. The same assertion holds for a semialgebraic function or arc approaching a finite boundary point.*

Reference. This is the ordinary semialgebraic curve selection lemma applied after compactifying $\mathbb{R}^N$, together with the Puiseux expansion theorem for one-variable semialgebraic germs; see, for example, [2, Sections 2.5 and 2.6] or [1, Chapter 3]. $\square$

4 The projective Puiseux descent certificate

Definition 4.1 (Projective Puiseux descent certificate). A projective Puiseux descent certificate for $g_* = -\infty$ is a tuple

$$Y(t) = (Y_0(t), Y_x(t)) \in \mathcal{P}^{n+1}$$

with the following properties.

(P1) $Y \in \widetilde{K}(\mathcal{P})$. Explicitly,

$$\tilde{c}_i(Y) = 0 \quad (i \in \mathcal{E}), \quad \tilde{c}_j(Y) \geq 0 \quad (j \in \mathcal{I}), \quad Y_0^2 + \|Y_x\|^2 = 1, \quad Y_0 \geq 0.$$

(P2) Y_0 has positive leading order:

$$Y_0(t) = bt^q + O(t^{q+\epsilon})$$

for some $b > 0$, $q > 0$, and $\epsilon > 0$.

(P3) The homogenized objective has a negative leading term whose order is too small to be cancelled by dehomogenization:

$$\tilde{g}(Y(t)) = -at^m + O(t^{m+\epsilon})$$

for some $a > 0$, $m \geq 0$, and $\epsilon > 0$, and

$$m < dq.$$

Theorem 4.2 (Soundness). *If a projective Puiseux descent certificate exists, then $g_* = -\infty$.*

Proof. By (P2), $Y_0(t) > 0$ for all sufficiently small positive t . Define

$$X(t) = \frac{Y_x(t)}{Y_0(t)}.$$

Since $Y \in \widetilde{K}(\mathcal{P})$, dehomogenization gives

$$c_i(X(t)) = Y_0(t)^{-\deg c_i} \tilde{c}_i(Y(t)) = 0 \quad (i \in \mathcal{E}),$$

while, because $Y_0(t)^{\deg c_j} > 0$,

$$c_j(X(t)) = Y_0(t)^{-\deg c_j} \tilde{c}_j(Y(t)) \geq 0 \quad (j \in \mathcal{I}).$$

Thus $X(t) \in K$ for all sufficiently small $t > 0$. Homogeneity gives

$$g(X(t)) = Y_0(t)^{-d} \tilde{g}(Y(t)).$$

Using (P2) and (P3),

$$g(X(t)) = -ab^{-d}t^{m-dq} + O(t^{m-dq+\epsilon'})$$

for some $\epsilon' > 0$. Since $m - dq < 0$, the right-hand side tends to $-\infty$. Therefore g is unbounded below on K . $\square$

Theorem 4.3 (Completeness). *Assume $K \neq \emptyset$. Then*

$$g_* = -\infty$$

if and only if a projective Puiseux descent certificate exists.

Proof. One direction is Theorem 4.2. Conversely, suppose $g_* = -\infty$. Since K is closed and g is continuous, g is bounded below on each compact set $K \cap \{\|x\| \leq R\}$. Hence any sequence in K along which $g \rightarrow -\infty$ must escape to infinity. By Lemma 3.1, after a Puiseux reparametrization there is an arc $\xi \in \mathcal{P}^n$ such that

$$\xi(t) \in K, \quad \|\xi(t)\| \rightarrow \infty, \quad g(\xi(t)) \rightarrow -\infty \quad (t \downarrow 0).$$

Define

$$Y_0(t) = (1 + \|\xi(t)\|^2)^{-1/2}, \quad Y_x(t) = Y_0(t)\xi(t),$$

where the positive square root is taken as a Puiseux germ. Then $Y_0 > 0$, $Y_0^2 + \|Y_x\|^2 = 1$, and, for each constraint,

$$\tilde{c}_i(Y_0, Y_x) = Y_0^{\deg c_i} c_i(\xi) = 0, \quad \tilde{c}_j(Y_0, Y_x) = Y_0^{\deg c_j} c_j(\xi) \geq 0.$$

Therefore $Y \in \widetilde{K}(\mathcal{P})$. Since $\|\xi(t)\| \rightarrow \infty$, the positive germ Y_0 tends to zero, so

$$Y_0(t) = bt^q + O(t^{q+\epsilon})$$

with $b > 0$ and $q > 0$.

Because $g(\xi(t)) \rightarrow -\infty$, its leading Puiseux term has negative coefficient and negative order:

$$g(\xi(t)) = -a't^{-\beta} + O(t^{-\beta+\epsilon})$$

with $a' > 0$ and $\beta > 0$. Hence

$$\tilde{g}(Y(t)) = Y_0(t)^d g(\xi(t)) = -a' b^d t^{dq-\beta} + O(t^{dq-\beta+\epsilon'}).$$

The components of Y are bounded Puiseux germs, so each has nonnegative order, and $\tilde{g}(Y)$, being a polynomial in those components, has nonnegative order unless it is identically zero. It is not identically zero because $Y_0^d g(\xi)$ is not identically zero and tends through negative values. Thus $m := dq - \beta \geq 0$. Since $\beta > 0$, we also have $m < dq$. This is exactly (P3), and the certificate is obtained. $\square$

Corollary 4.4 (The higher-order meaning when (4) fails). *Suppose (4) is infeasible and $g_* = -\infty$. Then there is a projective Puiseux descent certificate Y whose limit $(0, u) = \lim_{t \downarrow 0} Y(t)$ satisfies*

$$u \in K^\infty, \quad g_{\text{hom}}(u) = 0.$$

Along this branch the first negative term of $\tilde{g}(Y(t))$ occurs at positive order m with $0 < m < dq$.

Proof. The existence of Y follows from Theorem 4.3. Since $Y_0 \rightarrow 0$ and $Y \in \widetilde{K}$, the limit has the form $(0, u)$ with $u \in K^\infty$. By Proposition 2.1, infeasibility of (4) implies $g_{\text{hom}} \geq 0$ on K^∞ . If $g_{\text{hom}}(u) > 0$, then $\tilde{g}(Y(t)) \rightarrow g_{\text{hom}}(u) > 0$, contradicting the negative leading term in (P3). Hence $g_{\text{hom}}(u) = 0$. Since $\tilde{g}(Y(t)) \rightarrow g_{\text{hom}}(u) = 0$, the leading order m is positive. The strict inequality $m < dq$ is part of the certificate. $\square$

5 A Fritz–John tangency certificate

The projective certificate is exact but does not by itself say where to search. A standard way to focus the search is to minimize g on large spheres and use Fritz–John necessary conditions. This yields a second complete certificate supported on an explicit polynomial system.

Let $\mathcal{F}$ be the semialgebraic set of tuples

$$(x, \lambda_0, \lambda, \nu, \eta) \in \mathbb{R}^n \times \mathbb{R} \times \mathbb{R}^\mathcal{E} \times \mathbb{R}^\mathcal{I} \times \mathbb{R}$$

satisfying

$$c_i(x) = 0 \quad (i \in \mathcal{E}), \quad (5)$$

$$c_j(x) \geq 0, \quad \nu_j \geq 0, \quad \nu_j c_j(x) = 0 \quad (j \in \mathcal{I}), \quad (6)$$

$$\lambda_0 \geq 0, \quad \lambda_0^2 + \sum_{i \in \mathcal{E}} \lambda_i^2 + \sum_{j \in \mathcal{I}} \nu_j^2 + \eta^2 = 1, \quad (7)$$

$$\lambda_0 \nabla g(x) + \sum_{i \in \mathcal{E}} \lambda_i \nabla c_i(x) - \sum_{j \in \mathcal{I}} \nu_j \nabla c_j(x) - 2\eta x = 0. \quad (8)$$

The scalar η is the multiplier of the large sphere constraint; the sign convention is immaterial because η is free.

Definition 5.1 (Fritz–John Puiseux descent certificate). A Fritz–John Puiseux descent certificate is a tuple

$$(X, \Lambda_0, \Lambda, \mathcal{N}, H) \in \mathcal{P}^n \times \mathcal{P} \times \mathcal{P}^\mathcal{E} \times \mathcal{P}^\mathcal{I} \times \mathcal{P}$$

with the following properties.

$$(F1) \quad (X, \Lambda_0, \Lambda, \mathcal{N}, H) \in \mathcal{F}(\mathcal{P}).$$

(F2) X is unbounded as $t \downarrow 0$; equivalently, $\min_i \text{ord } X_i < 0$.

(F3) The objective has a negative leading term of negative order:

$$g(X(t)) = -at^{-\beta} + O(t^{-\beta+\epsilon})$$

for some $a > 0$, $\beta > 0$, and $\epsilon > 0$.

Theorem 5.2 (Soundness and completeness of the tangency certificate). *Assume $K \neq \emptyset$. Then $g_* = -\infty$ if and only if a Fritz–John Puiseux descent certificate exists.*

Proof. Soundness is immediate from (F1) and (F3): the conditions defining $\mathcal{F}$ include $X(t) \in K$ for all sufficiently small positive t , and (F3) gives $g(X(t)) \rightarrow -\infty$.

For completeness, assume $g_* = -\infty$. Let

$$\rho(K) = \{r > 0 : K \cap \{x : \|x\| = r\} \neq \emptyset\}.$$

This is an unbounded semialgebraic subset of $\mathbb{R}$, hence it contains a ray (R_0, ∞) . For $r > R_0$, define

$$\phi(r) = \min\{g(x) : x \in K, \|x\| = r\}.$$

The minimum exists because the feasible set is compact and nonempty. By Tarski–Seidenberg elimination, ϕ is a semialgebraic function. Since $g_* = -\infty$, the function ϕ is unbounded below along large radii. A one-variable semialgebraic function has a limit in $\mathbb{R} \cup \{\pm\infty\}$ as $r \rightarrow \infty$, so necessarily

$$\phi(r) \rightarrow -\infty.$$

By definable choice, after increasing R_0 if necessary, there is a semialgebraic selection $x(r) \in K \cap \{\|x\| = r\}$ such that $g(x(r)) = \phi(r)$. At each such r , the point $x(r)$ is a local minimizer of g under the equality constraints $c_i = 0$, the inequalities $c_j \geq 0$, and the sphere equality $\|x\|^2 - r^2 = 0$. The Fritz–John necessary conditions therefore give multipliers $(\lambda_0, \lambda, \nu, \eta)$, not all zero, with $\lambda_0 \geq 0$, $\nu_j \geq 0$, complementarity, and stationarity

$$\lambda_0 \nabla g(x(r)) + \sum_{i \in \mathcal{E}} \lambda_i \nabla c_i(x(r)) - \sum_{j \in \mathcal{I}} \nu_j \nabla c_j(x(r)) - 2\eta x(r) = 0.$$

After normalizing the multipliers to satisfy (7), another application of definable choice gives semialgebraic multiplier functions of r .

Set $t = 1/r$. Lemma 3.1 gives Puiseux expansions for the selected functions after shrinking to $0 < t < \epsilon$. Denote them by

$$(X(t), \Lambda_0(t), \Lambda(t), \mathcal{N}(t), H(t)).$$

They satisfy $(X, \Lambda_0, \Lambda, \mathcal{N}, H) \in \mathcal{F}(\mathcal{P})$, and $\|X(t)\| = 1/t$, so X is unbounded. Finally,

$$g(X(t)) = \phi(1/t) \rightarrow -\infty,$$

so the leading Puiseux term of $g(X(t))$ has negative coefficient and negative order. This is (F3). $\square$

6 Verification and search

For rational or algebraic input data, a branch of the polynomial system (5)–(8) can be encoded finitely by standard real-algebraic data. One convenient form is a primitive-element representation

$$P(t, \theta(t)) = 0, \quad X_k(t) = \frac{A_k(t, \theta(t))}{B_k(t, \theta(t))},$$

together with a Thom encoding or another isolating description of the real branch $\theta(t)$ for $0 < t < \epsilon$. Verification of a proposed certificate then reduces to polynomial arithmetic and univariate sign determination. In the common special case of a rational parametrization in $t^{1/q}$, one simply substitutes the parametrization into the constraints and objective, clears denominators, and proves the required signs on $(0, \epsilon)$, for instance by Sturm sequences.

The tangency theorem suggests the following search target.

1. Enumerate active sets $A \subseteq \mathcal{I}$. Impose $c_j(x) = 0$ for $j \in A$, impose $\nu_j = 0$ for $j \notin A$, and later certify $c_j(x) > 0$ for inactive inequalities.
2. Solve the polynomial equations consisting of the equalities, complementarity, multiplier normalization, and stationarity equations.
3. Compute or certify real branches of this system with $\|x\| \rightarrow \infty$.
4. Keep a branch if the inequality signs, multiplier signs, and negative leading term of $g(X(t))$ are certified.

By Theorem 5.2, this search target is complete. What remains outside the scope of this note is a complexity guarantee or an SDP hierarchy whose finite convergence would produce such a branch automatically in all cases. This is the precise sense in which the present result is partial progress on the “computationally convenient” part of Nie–Yang’s question.

7 The Motzkin example

Nie and Yang point out the following example, where the horizon moment ray is infeasible although the polynomial optimization problem is unbounded below:

$$\begin{aligned} \min \quad & g(x) = x_1 x_2 x_3 + x_1^2 x_2^2 (x_1^2 + x_2^2) + x_3^6 - 3x_1^2 x_2^2 x_3^2, \\ \text{s.t.} \quad & h(x) = x_1^2 + x_2^2 - 2x_3^2 = 0, \quad x_1 x_2 \geq 0. \end{aligned} \tag{9}$$

Here $d = 6$ and

$$g_{\text{hom}} = x_1^2 x_2^2 (x_1^2 + x_2^2) + x_3^6 - 3x_1^2 x_2^2 x_3^2$$

is the Motzkin form. It is nonnegative on $\mathbb{R}^3$, because

$$x_1^4 x_2^2 + x_1^2 x_2^4 + x_3^6 \geq 3|x_1^2 x_2^2 x_3^2| = 3x_1^2 x_2^2 x_3^2.$$

Therefore $\langle g_{\text{hom}}, z \rangle \geq 0$ for every $z \in R_6(K^\infty)$, and (4) is infeasible.

Nevertheless, set

$$X(t) = (t^{-1}, t^{-1}, -t^{-1}).$$

Then $h(X(t)) = 0$, $X_1(t)X_2(t) = t^{-2} > 0$, and

$$g(X(t)) = -t^{-3} \rightarrow -\infty.$$

The corresponding projective arc is

$$Y_0(t) = \frac{t}{\sqrt{t^2 + 3}}, \quad Y_x(t) = \frac{(1, 1, -1)}{\sqrt{t^2 + 3}}.$$

It satisfies $Y \in \widetilde{K}(\mathcal{P})$, and

$$\widetilde{g}(Y(t)) = Y_0(t)^6 g(X(t)) = -\frac{t^3}{(t^2 + 3)^3} = -\frac{1}{27}t^3 + O(t^5).$$

Thus $q = 1$, $m = 3$, and $dq = 6$, so $m < dq$. This is a projective Puiseux descent certificate. It detects exactly the phenomenon missed by (4): the order-zero term $g_{\text{hom}}(1, 1, -1)$ vanishes, while the lower-degree term $x_1 x_2 x_3$ gives the first negative projective term.

The same example also has a simple Fritz–John certificate. Write $s = t^{-1}$. Along $X = (s, s, -s)$,

$$\nabla g(X) = (-s^2, -s^2, s^2).$$

Take

$$\Lambda_0(t) = \frac{2t}{\sqrt{1 + 4t^2}}, \quad H(t) = -\frac{1}{\sqrt{1 + 4t^2}}, \quad \Lambda_h(t) = 0, \quad \mathcal{N}(t) = 0.$$

Then

$$\Lambda_0 \nabla g(X) - 2HX = 0, \quad \Lambda_0^2 + H^2 = 1,$$

and the equality, inequality, complementarity, and sign conditions are immediate. Therefore the arc is also a Fritz–John Puiseux descent certificate.

8 Conclusion

The horizon moment-ray condition (4) fails precisely when there is no horizon direction with $g_{\text{hom}} < 0$. This does not preclude unboundedness: lower-degree terms may become visible along branches on which the top homogeneous part cancels. The projective Puiseux certificate records that first negative term. It is sound and complete by a direct homogenization identity and semialgebraic curve selection at infinity. The Fritz–John variant shows that one can search among asymptotic critical branches of an explicit polynomial system.

Thus the result is a complete exact-algebraic characterization of unboundedness below, but only partial progress toward the strongest interpretation of Nie–Yang’s request for a computationally convenient certificate. The remaining challenge is to turn this higher-order branch certificate into a practically efficient moment/SOS or comparable computational framework.

References

- [1] Saugata Basu, Richard Pollack, and Marie-Françoise Roy. *Algorithms in Real Algebraic Geometry*. Springer, second edition, 2006.
- [2] Jacek Bochnak, Michel Coste, and Marie-Françoise Roy. *Real Algebraic Geometry*. Springer, 1998.
- [3] Theodore S. Motzkin. The arithmetic-geometric inequality. In Oved Shisha, editor, *Inequalities*, pages 205–224. Academic Press, 1967.
- [4] Jiawang Nie and Zi Yang. The multi-objective polynomial optimization. *Mathematics of Operations Research*, 49(4):2723–2748, 2024. doi:[10.1287/moor.2023.0200](https://doi.org/10.1287/moor.2023.0200). Available as arXiv:2108.04336.